

Fundamentals of Electricity

How can programmers not understand the basics of the machine before coding on it?



Let's start with the very basics then...

Electricity



Just some basic concepts for programmers

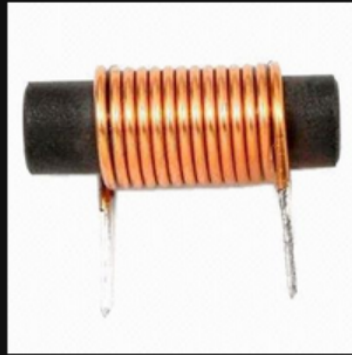
Electric Current

You cannot see it, but you know it's there...



Magnetism

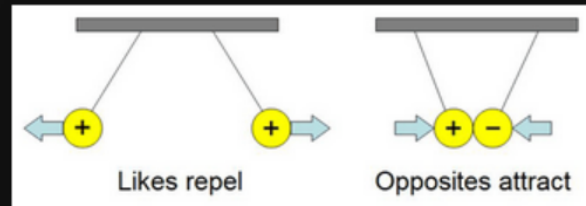
...and somehow related to magnetism!



magnets can induce electric current on a wire...

...electric current on a wire can generate a magnetic field

What does everybody know about magnets?



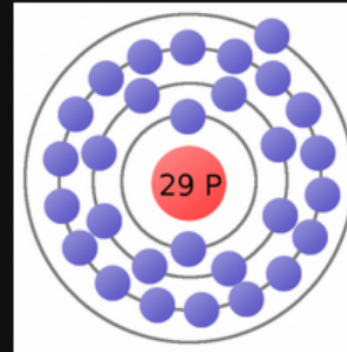
Equal charges (positive or negative) repel

Opposite charges attract

Bohr Model of the Copper Atom

- 29 protons (positive charges)
- 29 electrons in 4 shells (negative charges)
- The valence shell (last) has 1 electron on it

Charge attraction in Atoms works just like in Magnets



Why don't electrons collapse onto the protons?

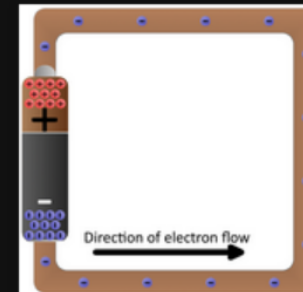
Electric Current

By providing adequate external force, that last and distant electron can jump from one atom to the next one, producing the phenomena we describe as electric current

- Positive pole has missing electrons
- Negative pole has excessive electrons

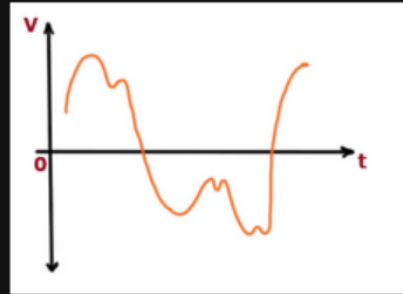
The charge difference between poles is called
Voltage

It basically means how hard the battery pushes electrons across the wire



Electric Signal

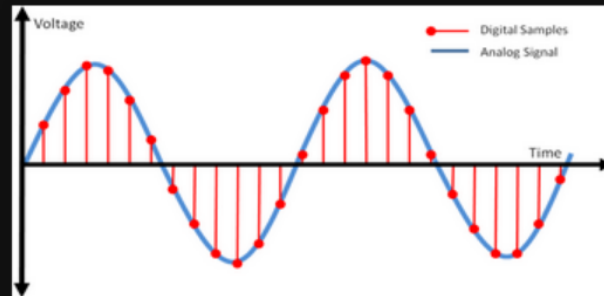
We can increase or decrease the current in a conductor by increasing or decreasing the voltage (push harder/softer)



Information (sound, audio, etc.) can be represented electrically by varying the current over time. We call this an Electric Signal

Analog Signal

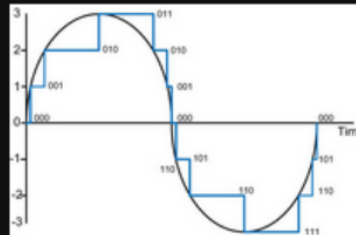
If the signal is continuous over time it is called Analog



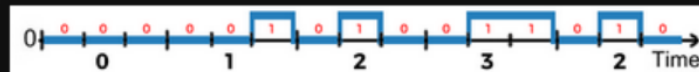
If we sample and work with specific discrete values only, the signal is called Digital

Digital Sampling

The discrete digital values are encoded in a set of 0s and 1s, representing the two possible voltage levels on a digital signal



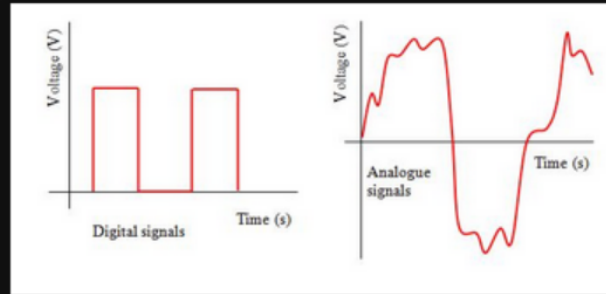
But what sort of coding scheme is this, capable of transforming decimal numbers into zeros and ones?



We need to understand some things about numbers before answering this question...

Digital Signal

Digital Signals have only two possible values of Voltage

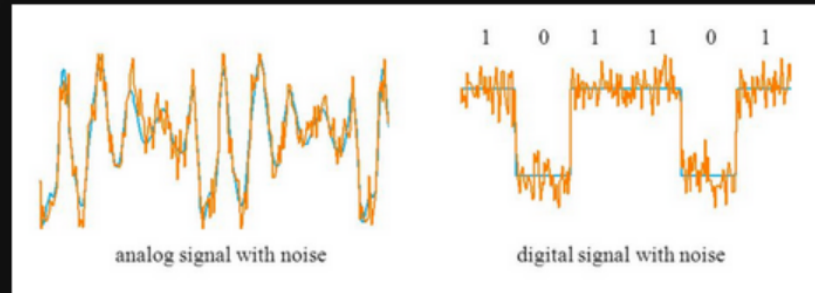
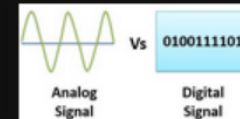


Common sense says Digital is better than Analog!

How can digital signals with only two values be superior to analog?

Digital vs Analog

The most important advantage of digital communication is immunity to noise interference



Even with significant amount of noise, it's easy to distinguish between a 0 and a 1

Electronic Circuits



Analog or digital, they are made of a combination of components allowing various simple and complex operations to be performed

Signals can be amplified, computations performed and data moved from one place to another



Electronic circuits are typically built on a Printed Circuit Board where all the components are connected using conductive tracks

But there is one electronic component which is very important...

Transistor

Analog signals can be amplified using semiconductor devices called Transistors. Electronic signals switch between voltage levels using Transistors.

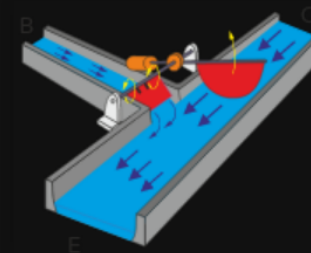
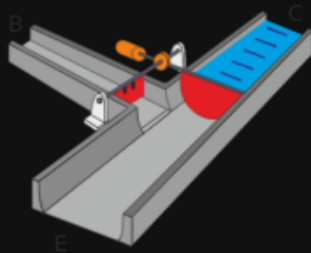
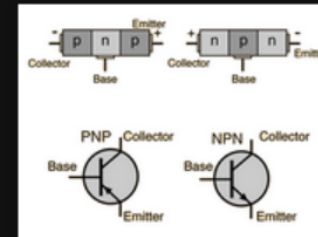


They are the fundamental building block of all modern electronic devices and millions of them can be found in modern systems

Transistor Operation

Switch the current on/off in a digital signal

- Made of silicon semiconductor material
- Fabricated with three separately doped regions
- Regions are called Base, Emitter and Collector
- P is positively doped
- N is negatively doped
- Small base current controls large emitter current



Integrated Circuits

An integrated circuit (also referred to as an IC, a chip, or a microchip) is a set of electronic circuits on one small flat piece of semiconductor material, normally silicon



As of 2019, the largest transistor count in a commercially available microprocessor is 39.54 billion MOSFETs

The CPU, Memory and most computer components are implemented as integrated circuits